

Data Mining Lab Project (10 Marks)

373CIS Data Mining

Objective

The objective of this project is to apply the complete data mining process to analyze a dataset and extract meaningful insights.

Project Workflow

You are required to perform the following steps:

1. Problem Definition
2. Data Understanding
3. Data Cleaning
4. Data Exploration
5. Data Transformation
6. Regression Analysis
7. Classification
8. Clustering
9. Text Analysis
10. Interpretation and Conclusion

Dataset Allocation

Each group is assigned a specific dataset. All groups must complete the full project tasks using their assigned dataset only.

Group Assignments

Group	Dataset Name	Description
Group 1	Banking Dataset	Customer financial behavior and loan approval
Group 2	Online Shopping Dataset	Customer purchases and transaction behavior
Group 3	Student Dataset	Academic performance and study habits
Group 4	Car Dataset	Vehicle performance and efficiency
Group 5	Employee Dataset	Employee performance and promotion decisions
Group 6	House Dataset	Property characteristics and pricing
Group 7	Hospital Dataset	Patient treatment and recovery outcomes
Group 8	Gym Dataset	Fitness activities and member progress
Group 9	Flight Dataset	Flight operations and customer satisfaction
Group 10	Restaurant Dataset	Orders, delivery performance, and customer ratings

Instructions

- Each group must work only on their assigned dataset.
- All project tasks (Problem Definition, Data Understanding, Cleaning, Exploration, Transformation, Regression, Classification, Clustering, Text Analysis, and Conclusion) must be completed.
- Groups must not exchange datasets.
- Each group should interpret results based on their dataset context.
- Ensure proper understanding of the dataset before applying any analysis.

1. Problem Definition (1 Mark)

Clearly define:

- The problem being addressed
- The objective of the analysis
- The expected outcome

Examples:

- Banking dataset: Predict loan approval and analyze spending behavior
- Student dataset: Analyze factors affecting student performance
- House dataset: Predict house prices
- Gym dataset: Understand fitness improvement

2. Data Understanding (1 Mark)

- Identify number of records and attributes
- Describe each attribute and its type (numeric, categorical, text)
- Understand dataset structure

3. Data Cleaning (2 Marks)

- Handle missing values
- Remove duplicate records
- Correct data types
- Ensure clean and structured dataset

Example:

- Replace missing values with suitable values
- Remove inconsistent records

4. Data Exploration (1 Marks)

- Use descriptive statistics
- Identify trends, patterns, and outliers
- Use visualizations to support findings

Examples:

- Banking: Income vs Expenditure
- Student: StudyHours vs Score
- Flight: Delay vs ServiceScore

5. Data Transformation (1 Mark)

- Aggregate data
- Create summarized views

Examples:

- Banking: Total spending per category
- Restaurant: Average rating per food type

6. Regression Analysis (1 Mark)

Requirement: Use a numerical variable as the target.

Targets for each dataset:

- Banking: Expenditure
- Shopping: PurchaseAmount
- Student: Score
- Car: Distance
- Employee: Salary
- House: Price
- Hospital: Cost
- Gym: FitnessScore
- Flight: ServiceScore
- Restaurant: Rating

Explain:

- Relationship between variables
- Strength of relationship

7. Classification (1 Mark)

Requirement: Use a categorical variable as the target.

Targets for each dataset:

- Banking: LoanStatus
- Shopping: PurchaseStatus
- Student: Result
- Car: Status
- Employee: Promotion
- House: Status
- Hospital: Status
- Gym: MembershipStatus

- Flight: Status
- Restaurant: OrderStatus

Explain:

- Factors influencing classification
- Model performance

8. Clustering (1 Mark)

- Use numerical variables only
- Group similar records into clusters

Examples:

- Banking: Income, Expenditure, Score
- Student: StudyHours, Score
- House: Size, Price
- Gym: Weight, FitnessScore

Explain:

- What each cluster represents
- Differences between clusters

9. Text Analysis (0.5 Mark)

- Use the Review column
- Identify sentiment (positive or negative)

Example:

- “Excellent” → Positive
- “Bad” → Negative

10. Interpretation and Conclusion (0.5 Mark)

- Summarize key findings
- Provide conclusions and recommendations

Examples:

- Banking: High-income customers spend more and are more likely approved
- Student: Study hours strongly influence performance
- Flight: Delays reduce customer satisfaction
- Restaurant: Faster delivery leads to higher ratings

Submission Rules

- Download and use the provided Word template to prepare your report.
- Complete all sections in the template, including results and interpretations.
- Ensure all tasks are clearly addressed and properly explained.
- Convert the final report to PDF format before submission.
- Submit the report as a single PDF file.
- The PDF must include:
 - Code
 - Outputs
 - Visualizations
 - Explanations
- Submit the R script file. It must be fully executable from top to bottom without errors.
- The R script must include:
 - Group information (group number and member names/IDs) at the top as comments
 - Clear comments indicating each task
 - Interpretations written as comments alongside the analysis

Notes:

- Both the PDF report and the R script must follow the naming convention below:
 - * **Group#_Section#.pdf**
 - * **Group#_Section#.R**
- Example: **Group10_Section2002.pdf** and **Group10_Section2002.R**

Academic Integrity

- Each group must complete the project independently using their assigned dataset.
- All work (code, analysis, and report) must be original.
- Each group must define its own business goals and provide its own interpretations.
- The following are considered cheating:
 - Copying from other groups
 - Sharing work between groups
 - Copying from the internet
 - Using AI tools to generate solutions
- Any cheating will result in a **zero (0) mark**.
- Late submissions are **not accepted** and will receive a **zero (0) mark**.

Marking Scheme

Task	Marks
Problem Definition	1
Data Understanding	1
Data Cleaning	2
Data Exploration	1
Data Transformation	1
Regression	1
Classification	1
Clustering	1
Text Analysis	0.5
Conclusion	0.5
Total	10